Remote Environmental Survey and Mapping of an Unseen Room

Stage 1 Concept and Feasibility Report — Work in Progress

Julian Joseph Amaro Project I (247-519-VA) Vanier College

October 30, 2026

Draft status. This document records the research and project information available so far. It is not a claim that the rover has passed its requirements. Version 1 and Version 2 subsystem tests on the exact watch demonstrated Wi-Fi scanning and display output. Version 2 physical photos show the touch-controlled SquareLine/LVGL interface in its searching state and displaying a scrollable result list; its source package also passed a fresh isolated Arduino CLI build using the documented target and dependencies. Rover communication, the survey map, summary and anomaly delivery, the 5 s requirement, and the full integrated build remain untested. The bill of materials and cost plan are intentionally excluded because they are separate Week 4 work.

1. Change Log

The submitted proposal remains the frozen baseline for needs N1–N8, watch constraints C1–C5, requirements REQ-RS-01–08, and risks R1–R3. The following additions make later project constraints explicit.

Item	Change and reason	Record
REQ-RS-09	Added the exact Waveshare ESP32-S3-Touch-AMOLED-2.06 as a required output that displays the map, summary, and anomaly indication.	L-02, L-03.1
REQ-RS-10	Added the final custom-PCB requirement. Stage 1 may still use development modules, breadboard, perfboard, or protoboard.	L-02, L-03.1
Gas interpretation	Clarified that the BME688 is a relative environmental/anomaly sensor, not selective toxic-gas detection or proof that a room is safe.	L-02–L-05, L-03.1

A Raspberry Pi Zero 2 W and a 3D-printed tracked tank are candidate components only. Neither is a frozen architecture decision or a verified compatible choice.

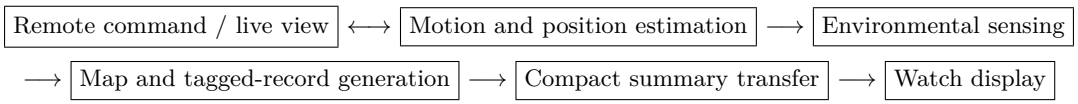
2. Concept Alternatives

Two genuinely different control approaches are under consideration.

	Alternative A: centralized control	Alternative B: divided control
Approach	One main computer coordinates video, mapping, sensing, communications, and motor commands through a motor driver.	A main computer handles video, mapping, storage, and communications, while a microcontroller handles motors, encoders, and time-sensitive sensor input.
Main uncertainty	Whether one processor and operating system can provide sufficiently predictable motor, encoder, video, and mapping behaviour together.	Whether the extra controller, firmware, interfaces, and synchronization are justified by improved real-time control.
Current status	Not scored; Raspberry Pi Zero 2 W remains a candidate.	Not scored; the microcontroller and interface are not selected.

The evaluation criteria and weights must be chosen and dated in the logbook before either alternative is scored. Likely criteria include requirement coverage, mapping risk, real-time control, implementation effort, power demand, integration complexity, and evidence obtainable during Project I. No winner is claimed yet.

Functional flow under investigation:



3. Feasibility Case

Claims begin at C6 because C1–C5 already identify watch constraints in the frozen proposal.

ID	Tests	Claim and present evidence	Class / source	Status
C6	RS-06, RS-09	The exact watch ran Version 1 and Version 2 Wi-Fi scans. V2 photos show the touch switch enabled, SEARCHING, FOUND 5, and a scrollable SSID/RSSI/channel list. Its preserved package passed a fresh isolated Arduino CLI build; matching hashes connect the tested original sketch, packaged sketch, and reference copy. The build used 1,223,571 bytes (93%) of its program partition and 112,992 bytes (34%) of global memory. A hidden access point does not broadcast its SSID, so passive scanning correctly displays `<hidden>`. The map, summary, anomaly display, rover connection, and 5 s transfer remain untested.	Measured V1/V2 tests; verified isolated V2 build; documented radio behaviour [1, 2]	**Partly supported**
C7	RS-03, RS-04	The BME688 documents temperature, humidity, pressure, and broad gas response. It can support a relative anomaly channel, but it is not selective or certified toxic-gas detection.	Documented research; L-02, L-03 [3]	**Partly supported**
C8	RS-04	Public BME688 projects demonstrate logging and experimental classification workflows, but no inspected public data set covered methane, propane/butane, CO, and H_2S together. No published result was reproduced.	Literature/repository review; L-05	**Open**
C9	RS-05	The tracked tank may provide a mechanical base, but its encoder integration, payload capacity, track slip, and mapping accuracy are not established for this project.	Documented/inferred; L-02	**Open**
C10	RS-01, RS-02, RS-05	A Pi Zero 2 W may be able to handle video and mapping, but processor load, thermal behaviour, motor-control timing, and power demand have not been calculated or measured for the final architecture.	Assumed; planned calculation and test	**Open**
C11	RS-01–RS-10	The integrated rover can meet the complete requirement set. No integrated hardware test exists yet.	Assumed; tests pending	**Open**

The research supports technical plausibility, not requirement compliance. The measured V1/V2 results support Wi-Fi scanning, display operation, and a touch-controlled result list; the isolated V2 build supports package reproducibility. They do not demonstrate the final survey screen or integrated rover link. No 2 m drive, video-latency test, mapping trial, BME688 bench test, watch connection or transfer timing, power test, or clean-copy one-command build of the integrated system has been completed. Vendor specifications remain component evidence, not measured integrated-system results.

4. Baselined Requirements

ID	Current requirement and reason	Status
REQ-RS-01	Complete a 2 m remotely commanded path with the operator outside the room; demonstrates remote entry.	Open: drive test pending.
REQ-RS-02	Display live phone video with latency at most 1 s; supports obstacle response.	Open: five timing trials pending.
REQ-RS-03	Map at least three tagged temperature, humidity, and relative-gas records; demonstrates spatial variation.	Partly supported: sensor capability documented; integration open.
REQ-RS-04	Flag a potential anomaly at 125% or more of baseline; this is a relative indicator, not a safety limit.	Partly supported: rule defined; implementation and sensor behaviour open.
REQ-RS-05	Map walls within ± 10 cm in at least four of five runs; preserves the proposal accuracy target.	Open: highest-priority mapping test pending.
REQ-RS-06	Deliver the completed summary to the watch within 5 s; provides portable results.	Partly supported: V1/V2 scanning and display measured; rover connection, summary transfer, and timing open.
REQ-RS-07	Keep total hardware cost at or below CAD $700; stakeholder constraint N7.	Open: cost work is separate and excluded here.
REQ-RS-08	Build firmware from a clean copy using one documented command; supports reproducibility.	Open: an isolated clean V2 subsystem build passed with the documented command; the full integrated build remains unverified.
REQ-RS-09	Use the exact Waveshare watch to display the map, summary, and anomaly indications.	Partly supported: V1/V2 display operation and V2 touch UI were measured; map, summary, anomaly display, and transfer remain open.
REQ-RS-10	Include a custom PCB in the final system; breadboard/modules alone are insufficient for the final build.	Open: binding plan; PCB not yet designed or built.

5. Risk Register

The following are the present research-backed risks. Probability and consequence scores are not assigned yet because the available measured evidence covers only the basic watch subsystem and does not support a defensible full-system ranking. The required $P = p \times c$ matrix remains open.

ID	Risk, assumption, and consequence	Evidence needed and mitigation
R1	Wheel slip and range noise may exceed the 10 cm error limit. Assumes odometry and ranging remain repeatable on the selected chassis. Failure would make wall and reading locations unreliable.	Five mapping runs against eight tape-measured points. Calibrate geometry, reduce speed, add encoders/ranging, or change mapping method.
R2	BME688 response may drift with humidity or be too slow for brief stops. Assumes a stable relative baseline. Failure could produce misleading anomaly indications.	Controlled safe baseline, warm-up, humidity, dwell-time, and repeatability tests. Keep claims relative and change the sensor or sampling method if needed.
R3	Doorway wireless traffic may delay phone video or the watch summary. Basic 2.4 GHz scanning works, but connection, coexistence under video load, and delivery timing are untested. Failure would affect REQ-RS-02 and REQ-RS-06.	Confirm a 2.4 GHz access point, then measure video and summary delivery together. Reduce data rate, buffer/resend, or test a phone-mediated path.
R4	The selected controller architecture may not meet processing, timing, thermal, or power needs.	Calculate the power/load budget and prototype the two highest-load functions before selection. Use divided control if centralized timing is inadequate.
R5	The final watch UI may exceed screen or program limits. V2 supports scrolling, but not the final survey layout, and its build already uses 93% of the program partition.	Test the map, summary, and anomaly screens with safe margins; track build size and reduce assets/features or adjust the approved partition plan if needed.

6. Gate 1 Readiness

The current research supports a **preliminary proceed-with-conditions** position. Measured V1/V2 tests on the exact watch demonstrated 2.4 GHz Wi-Fi scanning and AMOLED output. V2 photos show its touch-controlled SquareLine/LVGL result list, and a fresh isolated build verified the preserved package using the documented command and dependencies. This supports a reproducible watch subsystem, not connection to the rover, a survey map or summary, anomaly indications, delivery within 5 s, or the full integrated build. Its 93% program-partition use also leaves limited margin for the final interface. Research still shows that the BME688 cannot certify safety, the tracked tank is unproven for mapping, and the Pi Zero 2 W architecture is unselected.

The conditions are to implement and test the survey map, summary, and anomaly screens and transfer; measure watch delivery and video latency together; manage watch program size; and verify the full integrated clean-copy one-command build. The project must also fix and log the concept-selection criteria and weights, score the two architectures, select the controller structure, bench-test the BME688, measure mapping accuracy, and calculate the power/current budget. Until those steps are complete, every performance requirement remains open or partly supported.

References

[1] Waveshare, “ESP32-S3-Touch-AMOLED-2.06 Product Page and Wiki,” accessed October 23, 2026.

[2] Espressif Systems, “ESP32-S3 Series Datasheet” and “RF Coexistence Guide,” accessed October 23, 2026.

[3] Bosch Sensortec, “BME688 Gas Sensor Product Information,” accessed October 23, 2026.